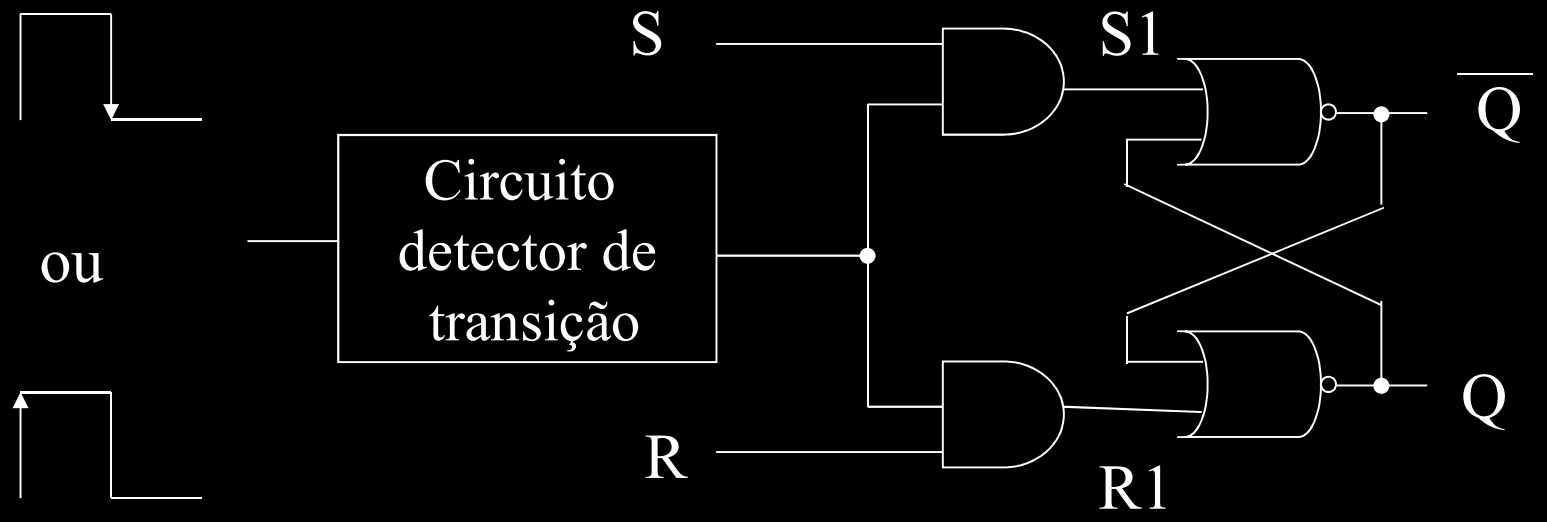
Circuitos Digitais I

Armazenar Dados

Flip Flop disparado por transição

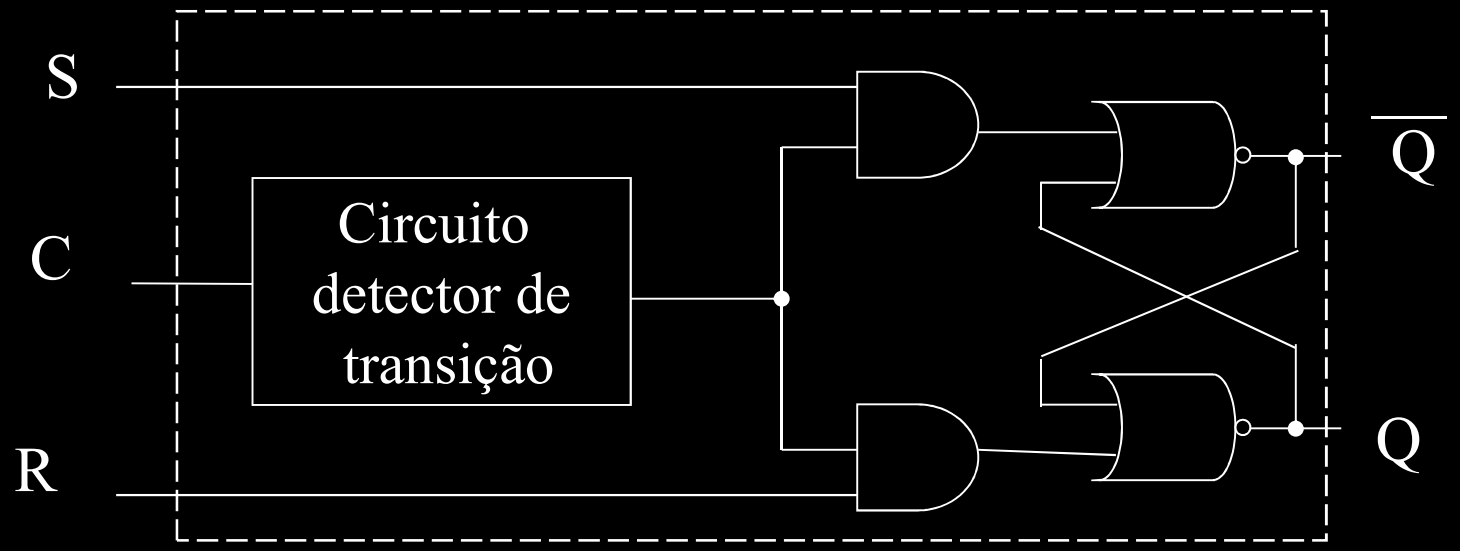
Clock:



Circuitos Digitais I

Armazenar Dados

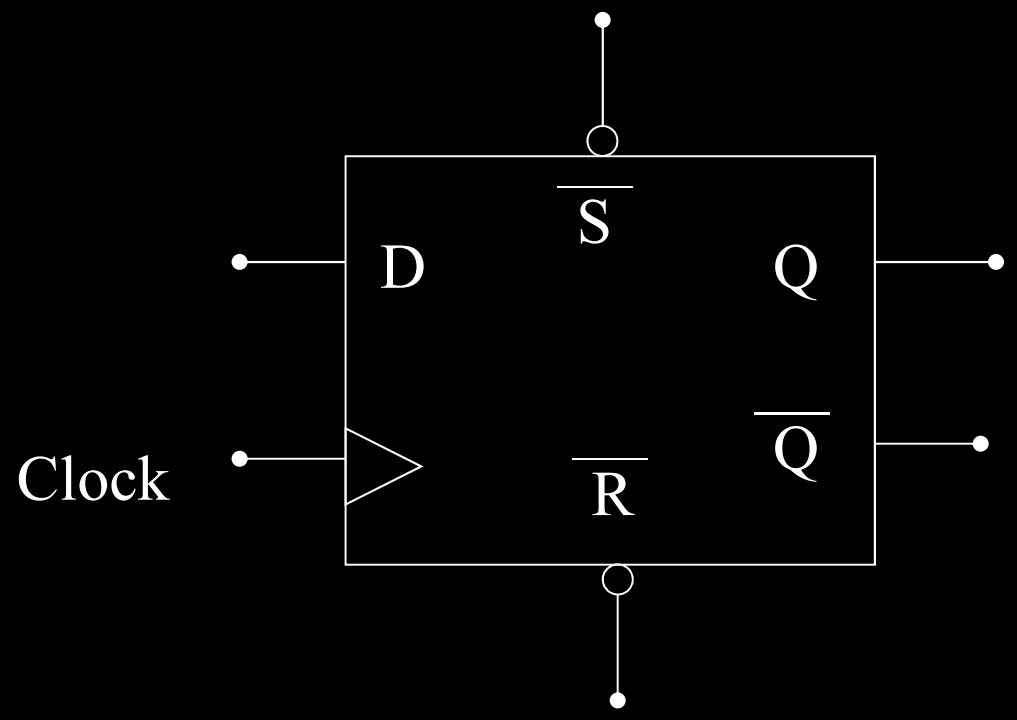
De forma geral os dispositivos de memória baseados em blocos lógicos são constituídos por SR Flip Flops



Circuitos Digitais I

Armazenar Dados

De forma geral os dispositivos de memória baseados em blocos lógicos são constituídos por SR Flip Flops



Circuitos Digitais I

Armazenar Dados

Componentes comerciais

National Semiconductor

February 1988

CD4013BM/CD4013BC Dual D Flip-Flop

General Description

The CD4013B dual D flip-flop is a monolithic complementary MOS (CMOS) integrated circuit constructed with N- and P-channel enhancement mode transistors. Each flip-flop has independent data, set, reset, and clock inputs and "Q" and "$\overline{Q}$" outputs. These devices can be used for shift register applications, and by connecting "$\overline{Q}$" output to the data input, for counter and toggle applications. The logic level present at the "D" input is transferred to the Q output during the positive-going transition of the clock pulse. Setting or resetting is independent of the clock and is accomplished by a high level on the set or reset line respectively.

Features

■ Wide supply voltage range	3.0V to 15V
■ High noise immunity	0.45 V_{DD} (typ.)
■ Low power TTL compatibility	fan out of 2 driving 74L or 1 driving 74LS

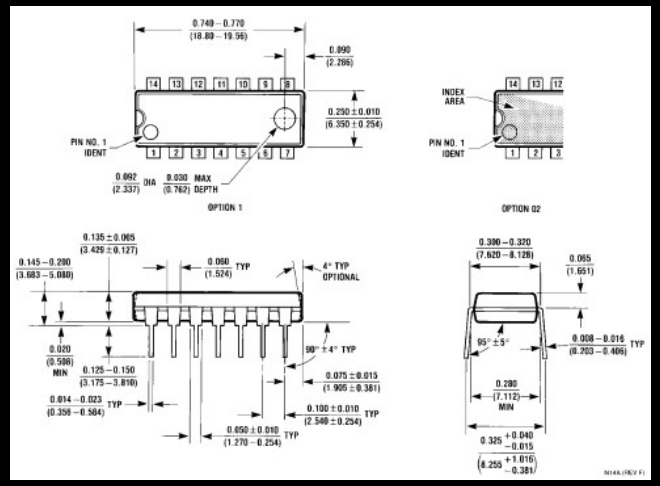
Applications

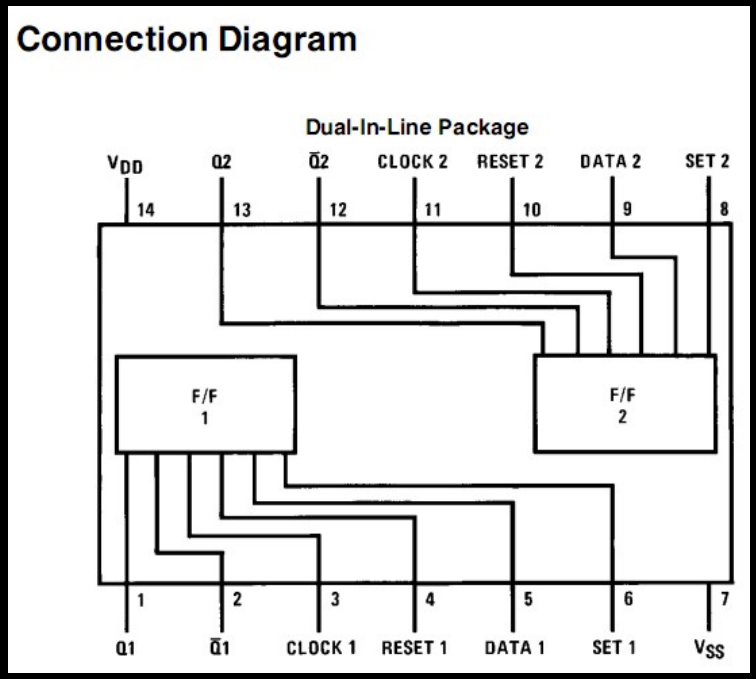
- Automotive
- Data terminals
- Instrumentation
- Medical electronics
- Alarm system
- Industrial electronics
- Remote metering
- Computers

Circuitos Digitais I

Armazenar Dados

Componentes comerciais





CL†	D	R	S	Q	$\overline{Q}$
↗	0	0	0	0	1
↗	1	0	0	1	0
↘	x	0	0	Q	$\overline{Q}$
x	x	1	0	0	1
x	x	0	1	1	0
x	x	1	1	1	1

No change
† = Level change
x = Don't care case

Circuitos Digitais I

Armazenar Dados

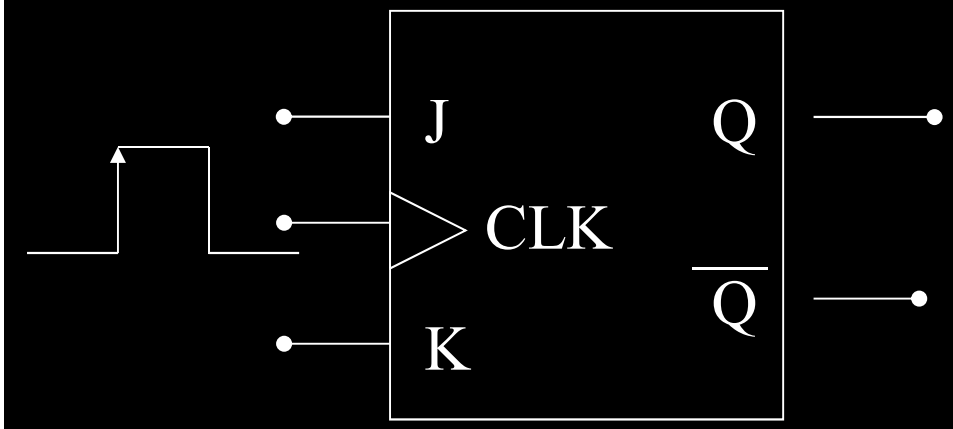
Flip Flop J-K com clock

- Disparado por transição do clock
- Controle das entradas da mesma forma que o Flip Flop SRC
- Diferente do SRC , Flip Flop J-K não resulta em estado inválido quando J=K=1

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock



J	K	CLK	Q
0	0	↑	Q
1	0	↑	1
0	1	↑	0
1	1	↑	Comuta

Vai para o
estado oposto

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

Aprimora FF SR ao interpretar a condiçã S=R= 1 como

Inversão (Toogle)

0	1	↑	0
1	1	↑	Comuta

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

J = 1, K = 0 é um comando para ativar (set) a saída do flip-flop

J	K	CLK	Q
0	0	↑	Q
1	0	↑	1
0	1	↑	0
1	1	↑	Comuta

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

J = 0, K = 1 é um comando para desativar (reset) a saída do flip-flop alternando a condição inicial

J	K	CLK	Q
0	0	↑	Q
1	0	↑	1
0	1	↑	0
1	1	↑	Comuta

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

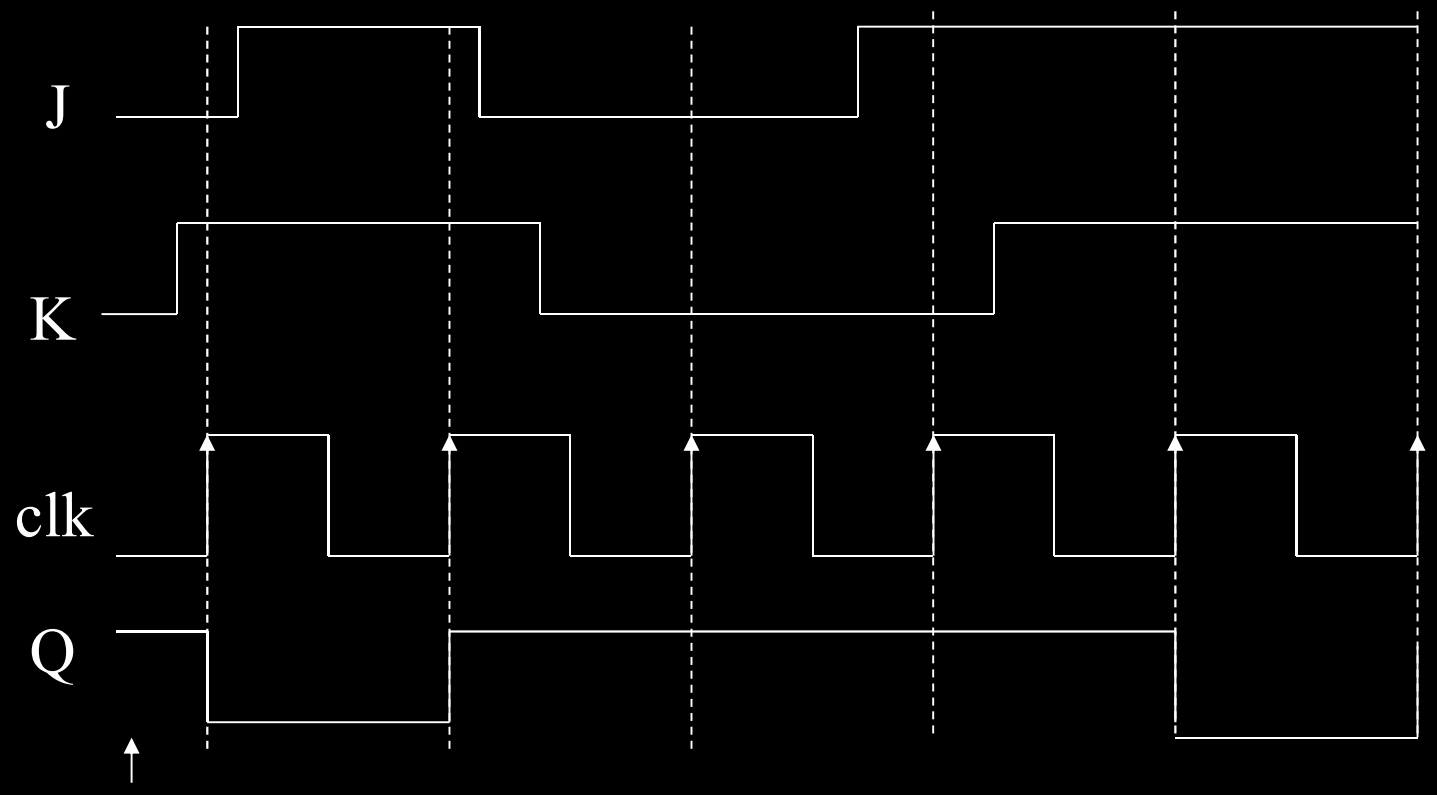
J = K = 1 é um comando para inverter o flip-flop, trocando o sinal de saída

J	K	CLK	Q
0	0	↑	Q
1	0	↑	1
0	1	↑	0
1	1	↑	Comuta

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

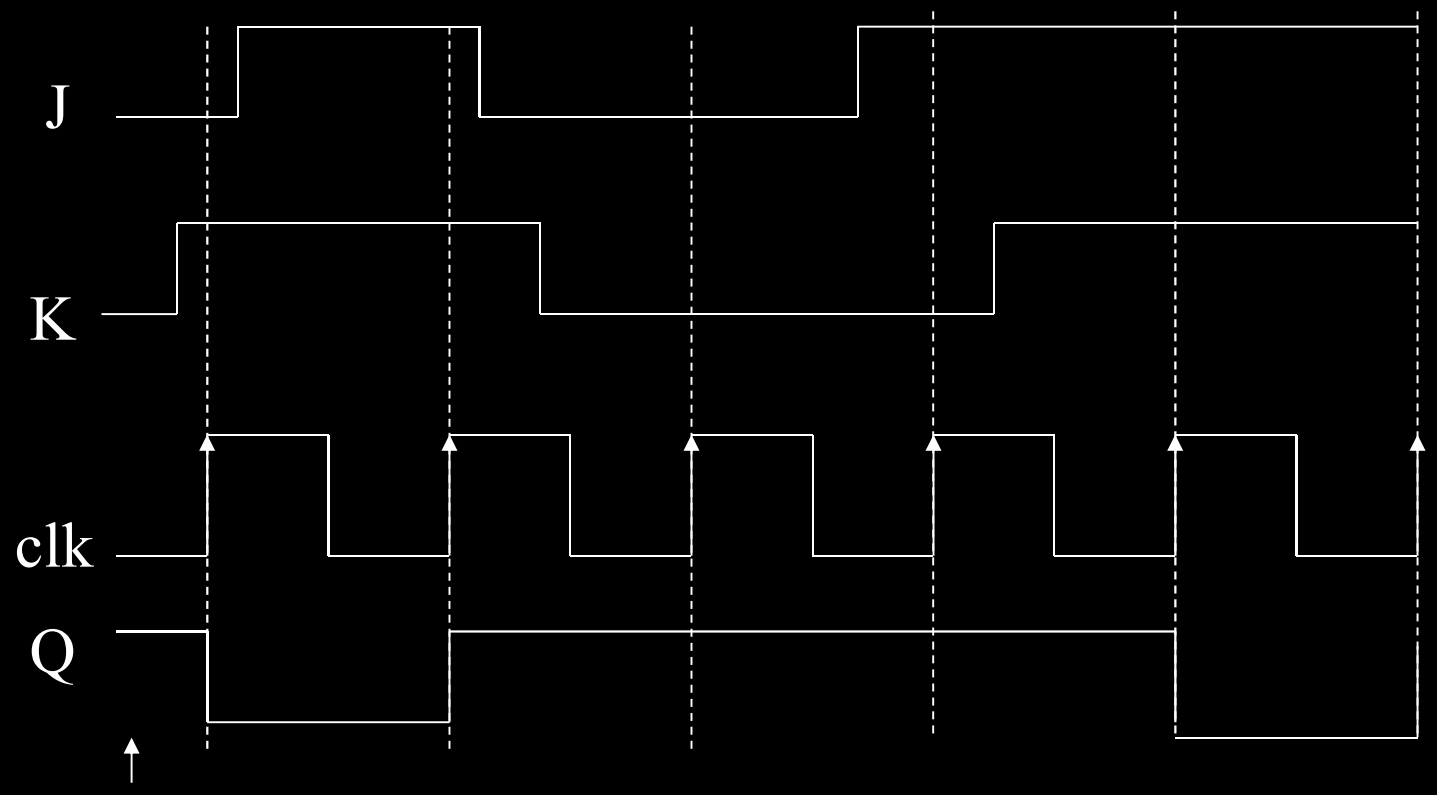


Estado inicial: entradas em 0 e Q = 1

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

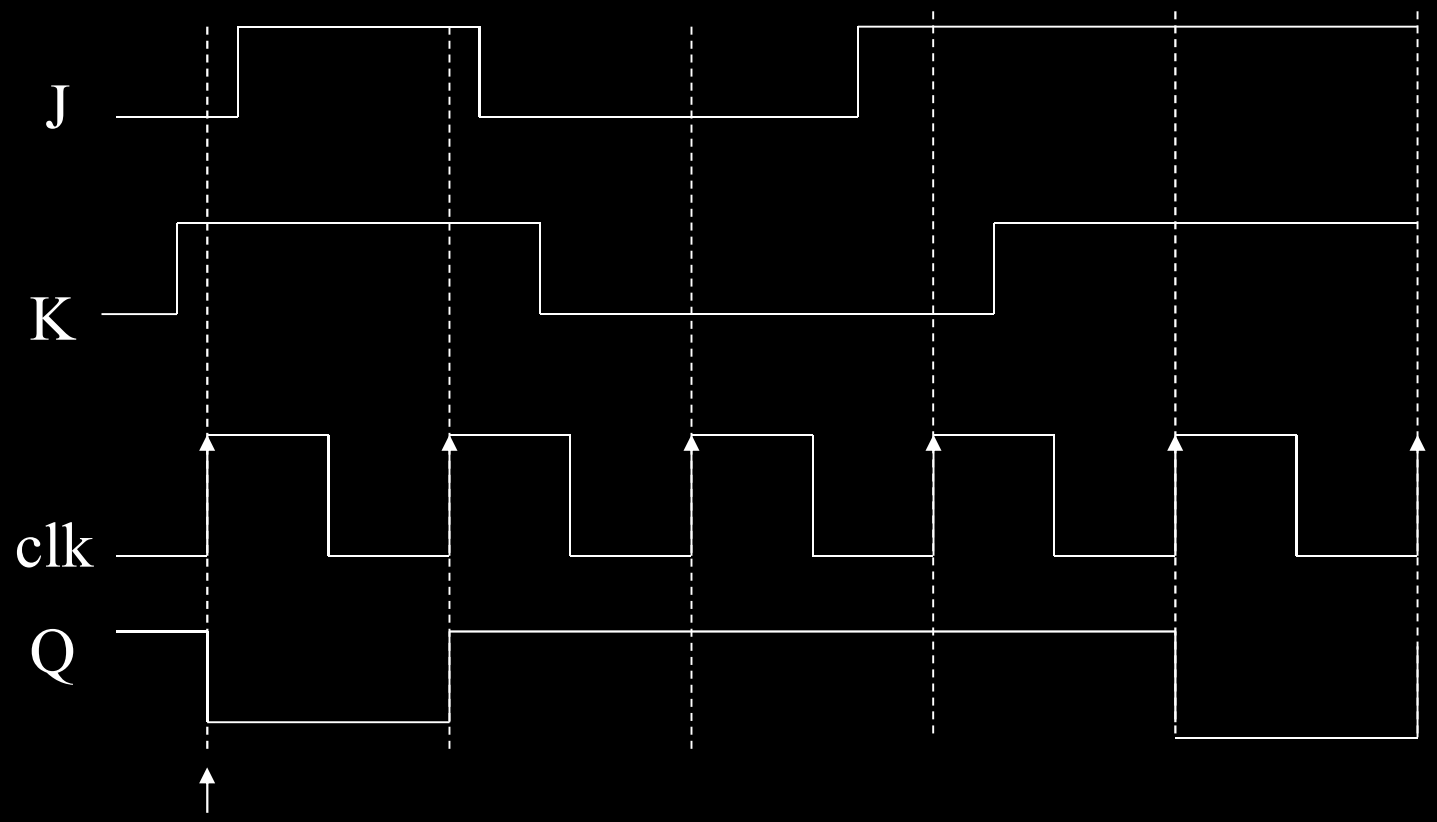


Estado inicial: entradas em 0 e Q = 1

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

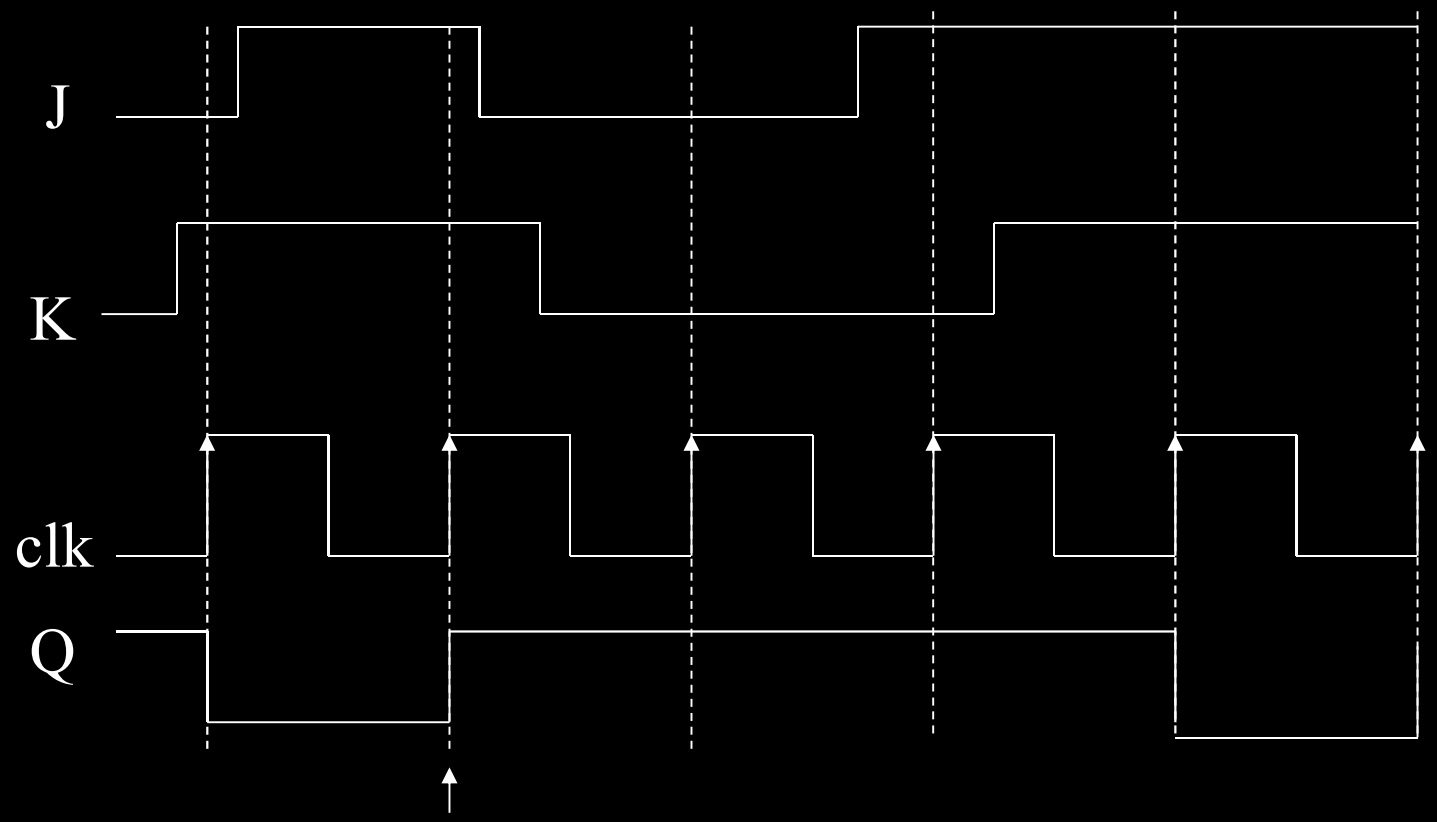


Transição 1: J = 0, K = 1 o FF resetado portanto Q =0

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

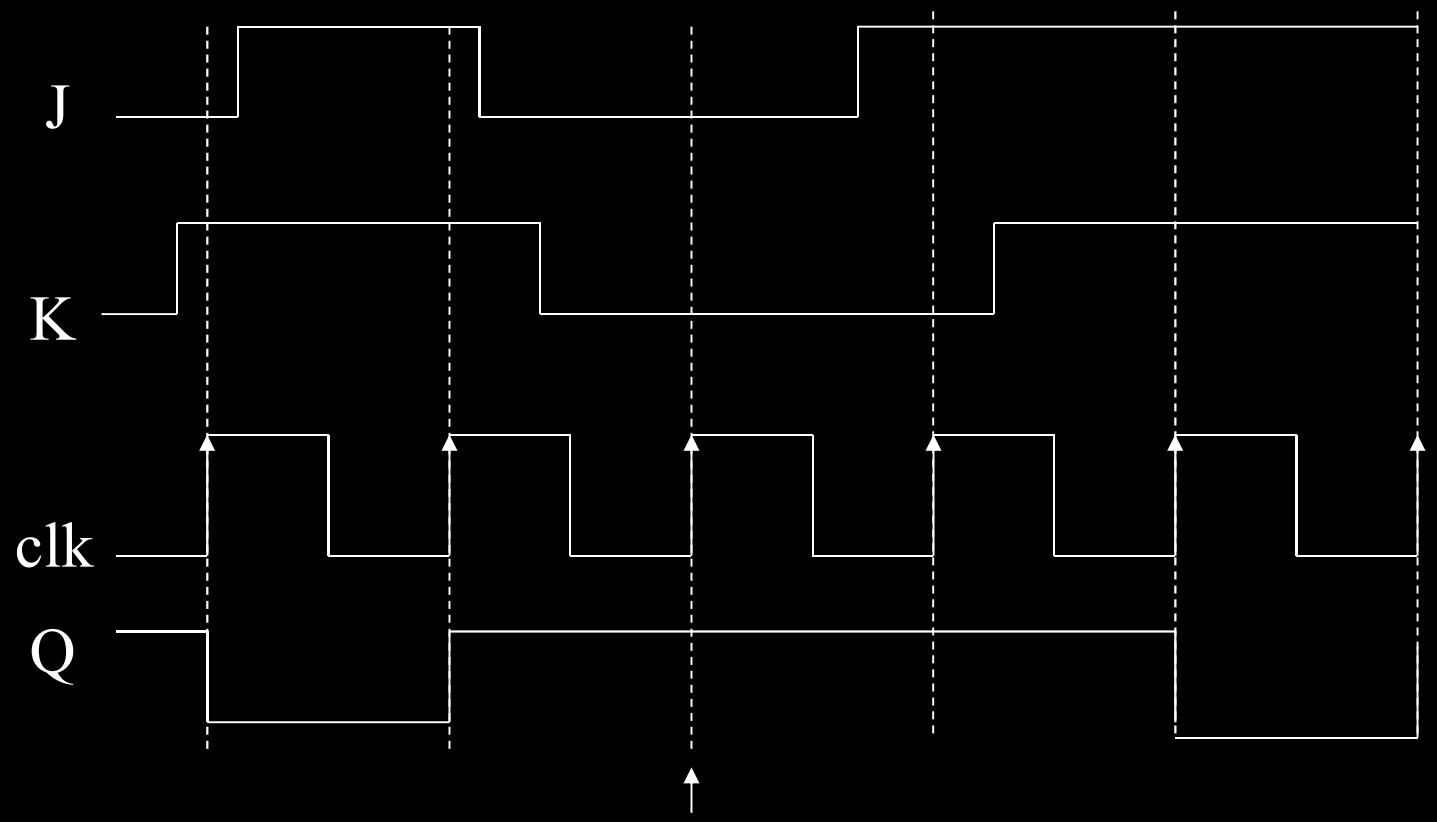


Transição 2: J = 1, K = 1 o FF comuta portanto Q =1

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

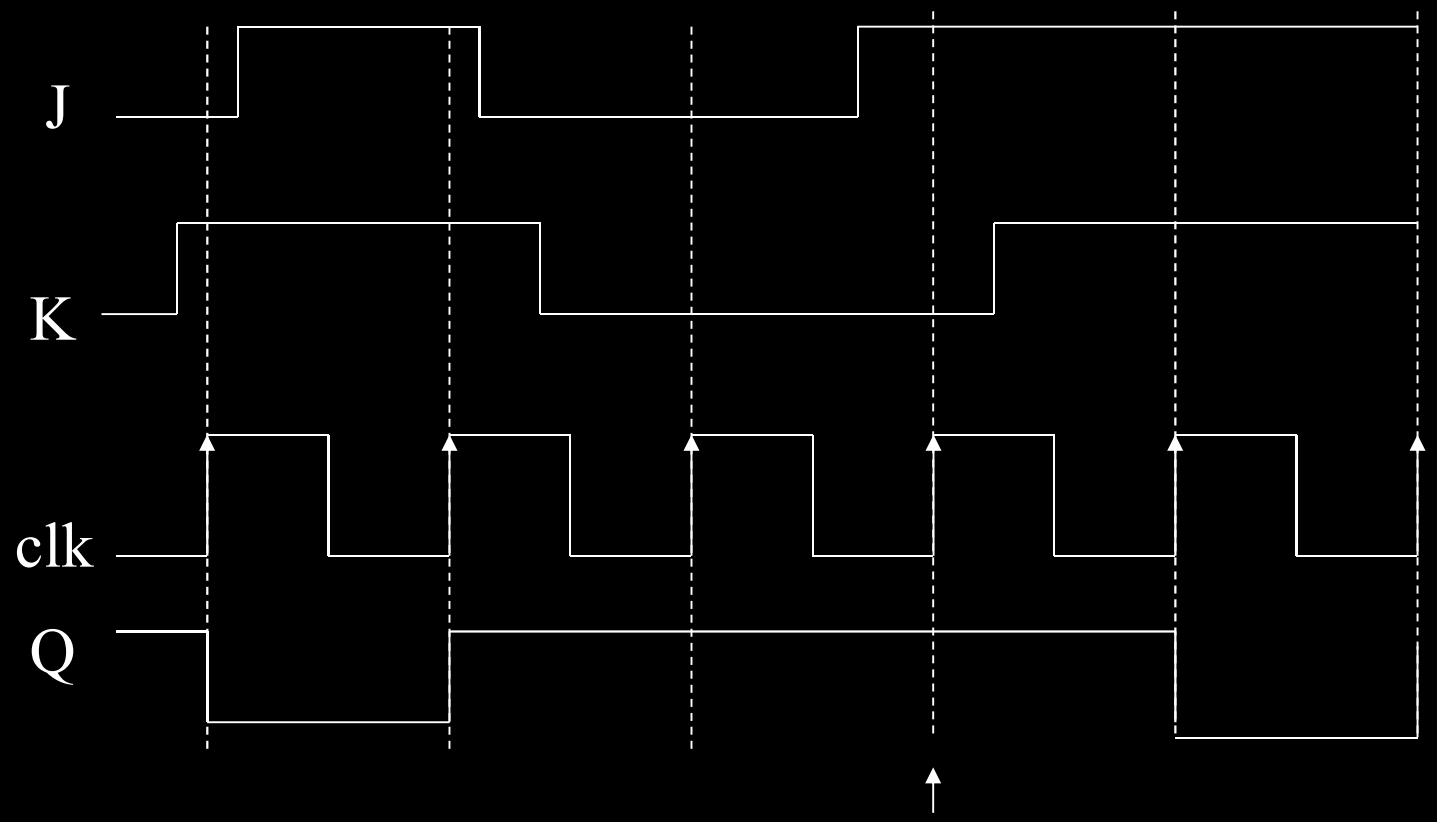


Transição 3: J = 0, K = 0 o FF não muda portanto Q =1

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock

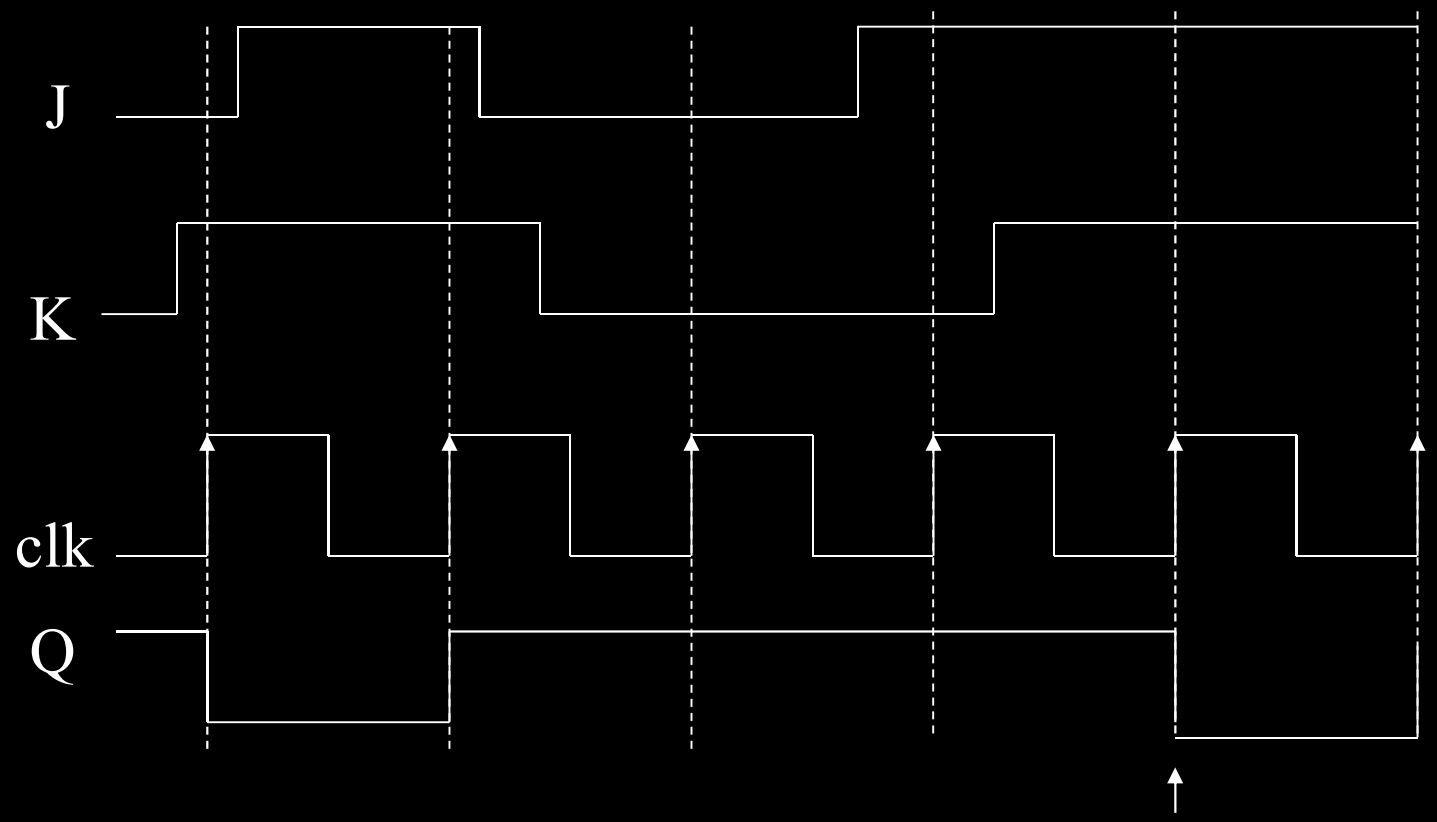


Transição 4: J = 1, K = 0 o FF vai a 1 portanto Q =1

Circuitos Digitais I

Armazenar Dados

Flip Flop J-K com clock



Transição 5: J = 1, K = 1 o FF comuta portanto Q =0

Circuitos Digitais I

Armazenar Dados > Flip Flop J-K com clock > VHDL

```
entity JK_FF is
PORT( J,K,CLOCK: in std_logic;
Q, QB: out std_logic);
end JK_FF;
```

Circuitos Digitais I

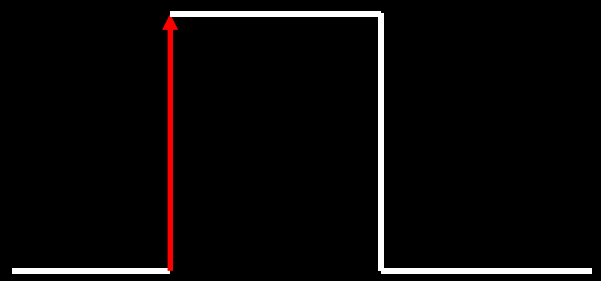
Armazenar Dados > Flip Flop J-K com clock > VHDL

```
Architecture behavioral of JK_FF is
begin
PROCESS(CLOCK)
variable TMP: std_logic;
begin
if(CLOCK='1' and CLOCK'EVENT) then
   if(J='0' and K='0')then
     TMP:=TMP;
   elsif(J='1' and K='1')then
     TMP:= not TMP;
   elsif(J='0' and K='1')then
     TMP:='0';
   else
     TMP:='1';
  end if;
end if;
Q<=TMP;
Q <=not TMP;
end PROCESS;
end behavioral;
```

Circuitos Digitais I

Armazenar Dados > Reset Assíncrono e Síncrono

Especificação VHDL 93
(Recomendado)

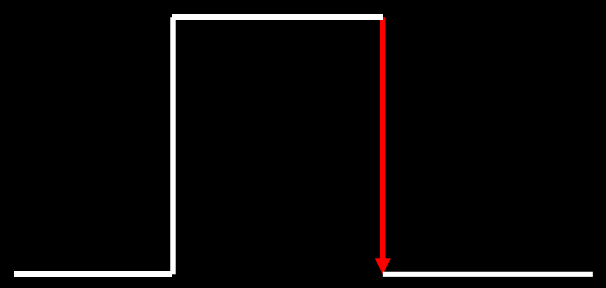


```
if Rising_edge(clock) then
```

```
if(CLOCK='1' and CLOCK'EVENT) then
```

Circuitos Digitais I

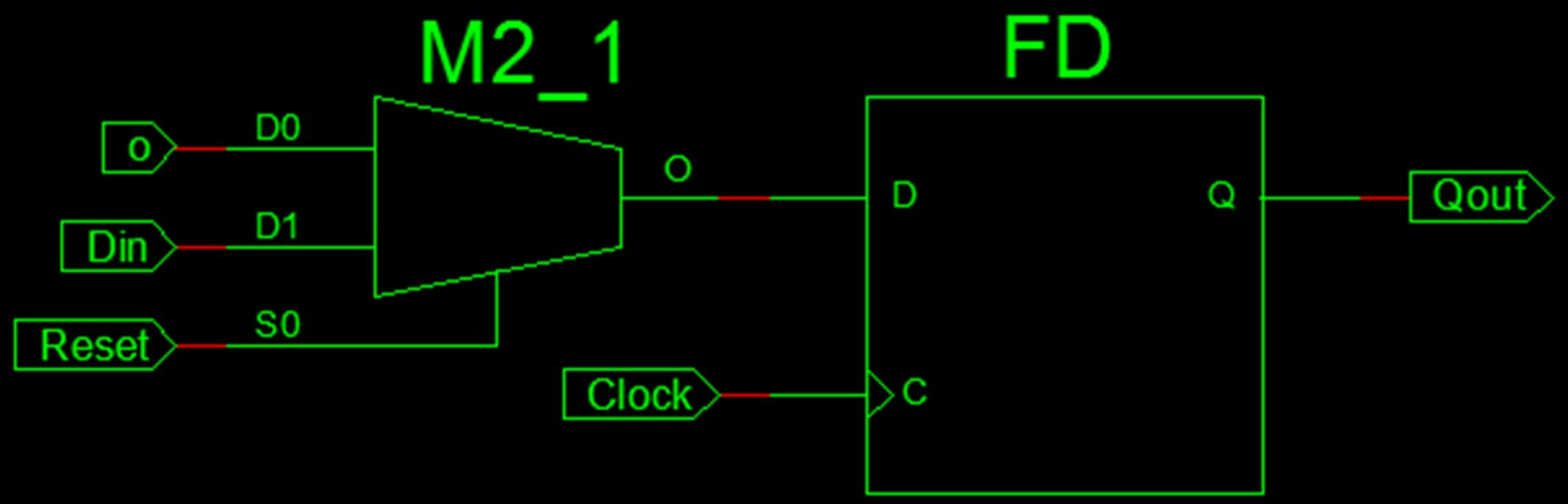
Armazenar Dados > Reset Assíncrono e Síncrono



```
if falling_edge(clock) then
```

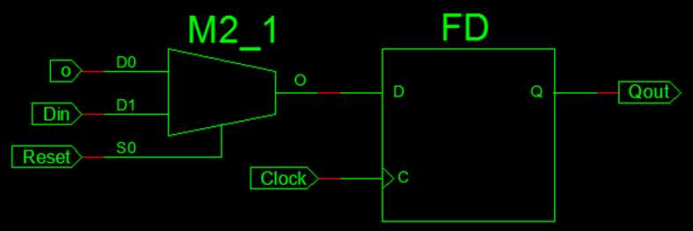
Circuitos Digitais I

Armazenar Dados > Reset Assíncrono



Circuitos Digitais I

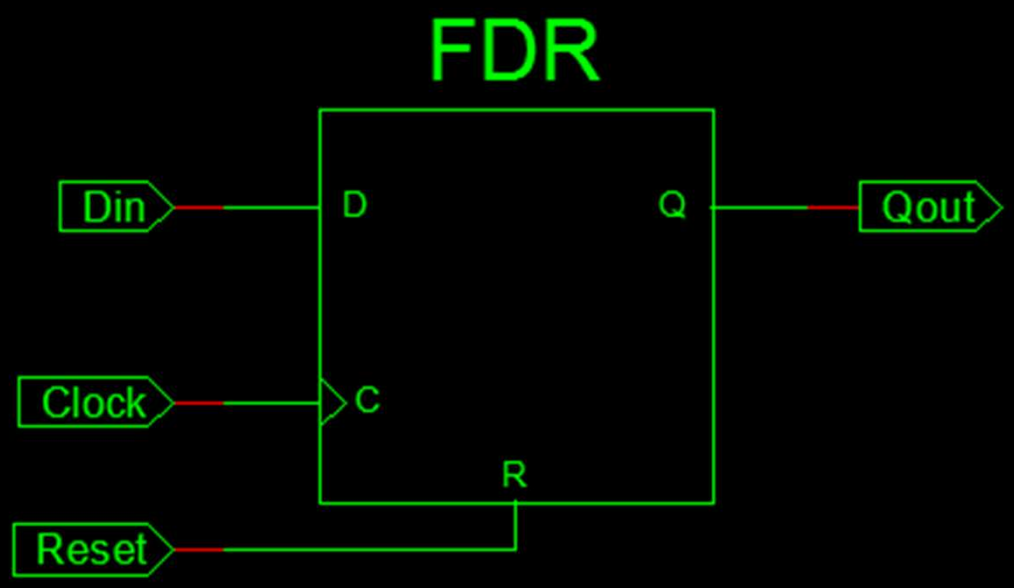
Armazenar Dados > Reset Assíncrono




```
entity asynchronous_reset is
port(clock, Reset, data: Std_logic;
     q: srd_logic);
end asynchronous_reset;
architecture bhv of asynchronous_reset is
begin
process(clock,Reset,data)
  if Reset = '1' then
    q <= '0';
  elsif Rising_edge(clock) then
    q <= data;
  end if;
end process;
end bhv;
```

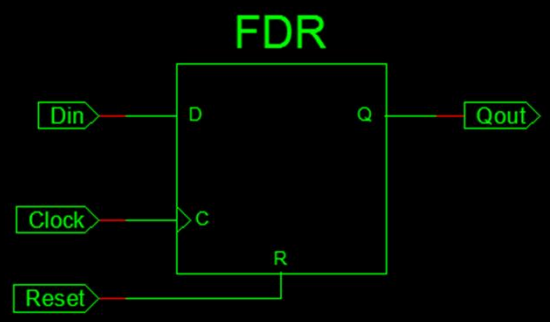
Circuitos Digitais I

Armazenar Dados > Reset Síncrono



Circuitos Digitais I

Armazenar Dados > Reset Síncrono

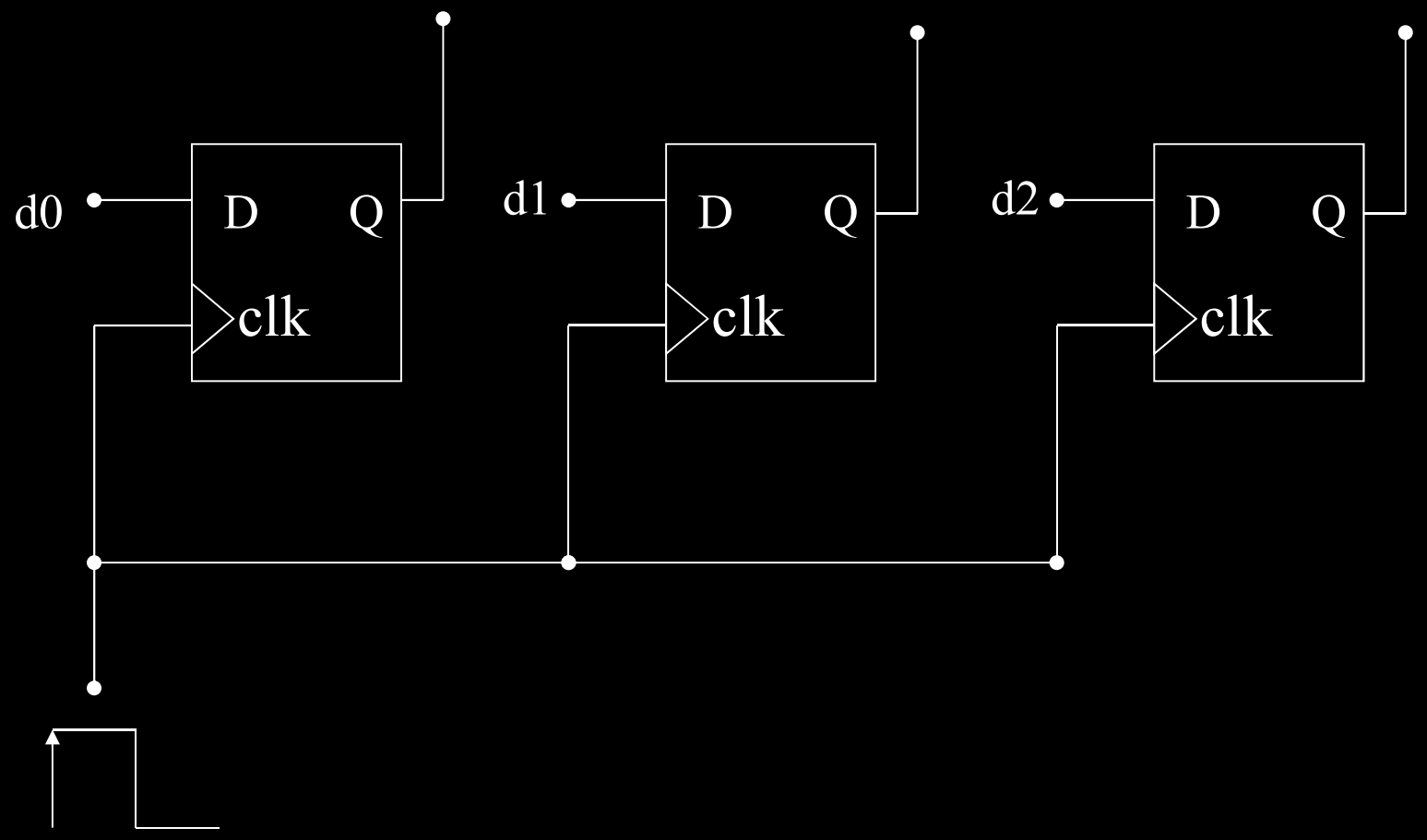



```
entity synchronous_reset is
port(clock, Reset, data: Std_logic;
     q: std_logic);
end synchronous_reset;
architecture bhv of synchronous_reset is
begin
process(clock,Reset,data)
  if Rising_edge(clock) then
     if Reset = '0' then
        q <= '0';
     else
        q <= data;
     end if;
  end if;
end process;
end bhv;
```

Circuitos Digitais I

Armazenar Dados

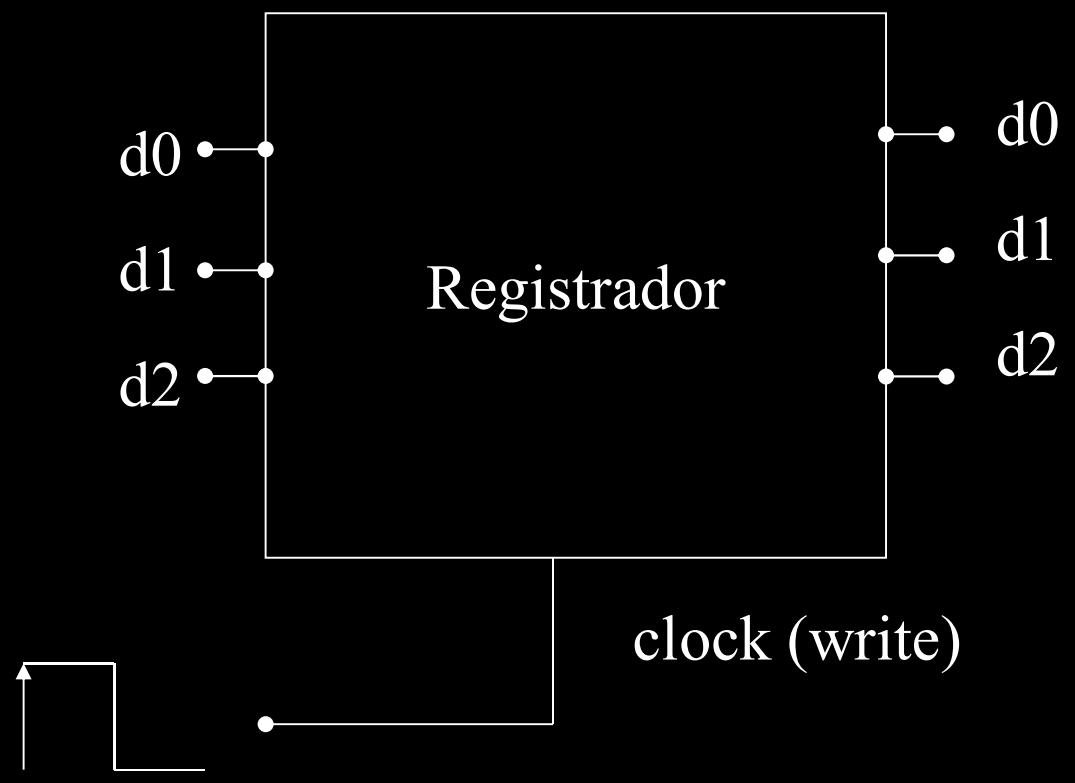
Registradores



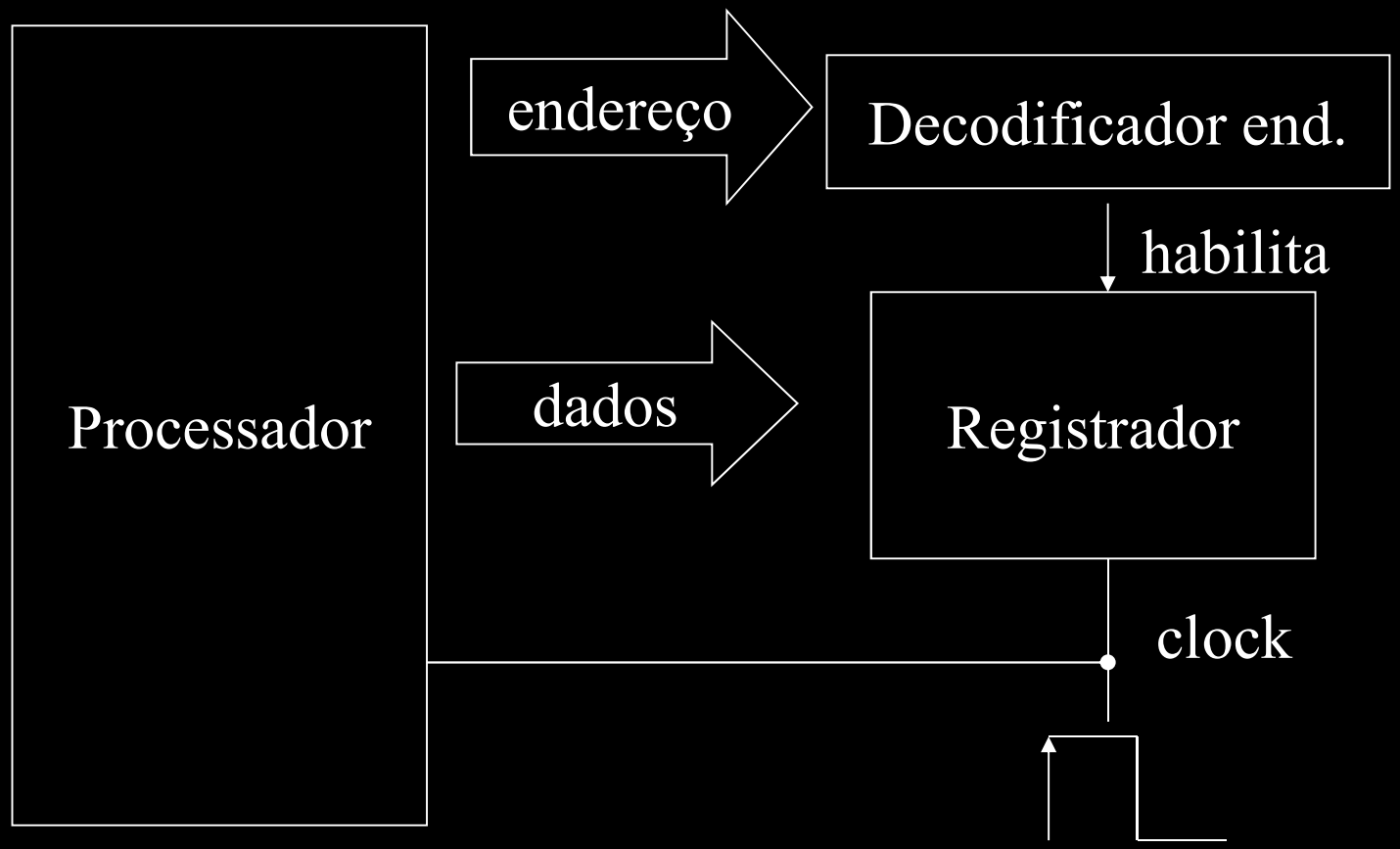
Circuitos Digitais I

Armazenar Dados

Registradores



Circuitos Digitais I

Armazenar Dados

Registradores: aplicação